ODD 2026

Database Management Systems Lab (24B15CS213)

Practice Lab 1

Question 1 A Library Management System stores information about Books, Members, Authors, and Book Issue Transactions.

Identify:

- The entities involved.
- At least four attributes for each entity.
- The primary key for each entity.

Question 2 Using any CASE Tool (MySQL Workbench, Draw.io, Lucidchart, ERDPlus, Oracle SQL Developer Data Modeler, etc.), create an Entity-Relationship (ER) Diagram for the Library Management System.

Your ER diagram should include:

- Entities
- Attributes
- Relationships
- Cardinality (1:1, 1:M, M:N)
- Primary Keys
- Foreign Keys

Question 3 Based on the ER diagram created in Question 2, explain how the following constraints are applied:

a) Entity Integrity Constraint

b) Referential Integrity Constraint

c) Key Constraints

d) Domain Constraints

Support your answer with suitable examples from your ER diagram.

Question 4 Consider the following modifications to the Library Management System:

- A book can have multiple authors.
- A member can reserve a book before it is issued.
- Each reservation has a reservation date and status.

Update your ER model by:

- Identifying any new entities or relationships.
- Modifying the ER diagram accordingly.
- Briefly justify the changes made

Question 5 A student has designed an ER diagram for an Online Shopping System with the following assumptions:

- A customer can place multiple orders.
- An order contains multiple products.
- A product belongs to exactly one supplier.
- The Order entity has no primary key.
- The Customer entity contains an attribute named Age, which is updated every year manually.

Tasks:

1. Identify any three design errors or inconsistencies in the above ER model.
2. Suggest appropriate corrections with justification.
3. State the type of constraint (entity integrity, referential integrity, key constraint, or domain constraint) affected by each error.

Question 6 Consider the following two real-world scenarios:

- Scenario A: A Hospital Management System where a doctor can treat many patients, and a patient can be treated by many doctors.
- Scenario B: A Banking System where each bank branch manages multiple accounts, but each account belongs to only one branch.

Tasks:

1. Draw separate ER diagrams for both scenarios using a CASE tool.
2. Compare the relationship types and cardinalities used in each model.
3. Explain how entity integrity and referential integrity are maintained in both schemas.